# Network Programming (IE2102)

2nd Year, 1st Semester

# Assignment

## PART A – Take-Home Component

Submitted to

Sri Lanka Institute of Information Technology

IT24103920

Kithmina H.M.O.T

In partial fulfillment of the requirements for the

Bachelor of Science Special Honors Degree in Information Technology

3/31/2026

***Important:***

*Remember*

- *Plagiarism is considered as a serious crime against intellectual property in academic environments.*
- *By including the declaration page mentioned below, you abide by the laws of the institute to accept 0 marks for the report, in case you get caught attempting plagiarism.*

**Annexure 1**

## General Formatting

1. Paper size: A4
2. Contents font & size: Times New Roman, size 12
3. Chapter headings/sub headings font & size: Times New Roman Bold, size 12
4. Page numbering:
   - Cover page should not contain any number
   - Declaration page to table of figures page should be numbered using Roman numbers at bottom center of the page
   - Content pages, references page(s) and annexes (if any) should be numbered using Arabic numbers at the bottom center of the page
5. Margins: 1 inch margins on left, right, top and bottom
6. Line spacing: 1 line spacing

## Figure Numbering:

1. Each figure should be numbered at bottom center of the figure along with a figure name.
2. Figure number should start with word 'Figure' followed by the 'main chapter number' and 'figure number within the chapter' separated by a 'period sign (.)' and followed by a 'colon (:)'.

For example,

Figure 2.6: XYZ Connector

## Declaration Page:

Declaration page should contain the following. Reports that does not contain this declaration 'AS IS' will not be accepted as valid reports and will be awarded 0 marks.

### Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

## 1. Introduction

This report documents the design, implementation, and testing of a secure multiprocessor TCP application server developed for the IE2102 Network Programming module. The server is implemented in C and the client in Python, fulfilling all requirements of Part A of the assignment.

The system demonstrates custom TCP protocol framing, multiprocessing using fork(), authentication with hashed passwords, abuse protection, and persistent audit logging.

**Registration Number:** IT24103920

**Server Port:** 50920 (PORT = 50 + last 3 digits of IT24103920)

**Server ID:** SID:1039 (middle 4 digits of IT24103920)

**Log File:** server_IT24103920.log

## 2. System Overview

The system consists of two components: a TCP server (server_3920.c) and a Python client (client_3920.py). The server listens on port 50920, forks a child process for each incoming client, and handles REGISTER, LOGIN, and LOGOUT commands over a custom length-prefixed protocol.

| File | Language | Role | Port |
|---|---|---|---|
| server_3920.c | C | TCP Server | 50920 |
| client_3920.py | Python | TCP Client | 127.0.0.1 |

## 3. Evidence Screenshots

*All screenshots below were captured during a live demonstration of the running system. Each screenshot visibly shows the registration number IT24103920 in the terminal prompt or output.*

### 3a. Hostname Configuration

Command used: hostname

*This screenshot shows the system hostname*

### Screenshot 3a – Hostname Configuration

*Run: hostname | Screenshot must show system hostname*

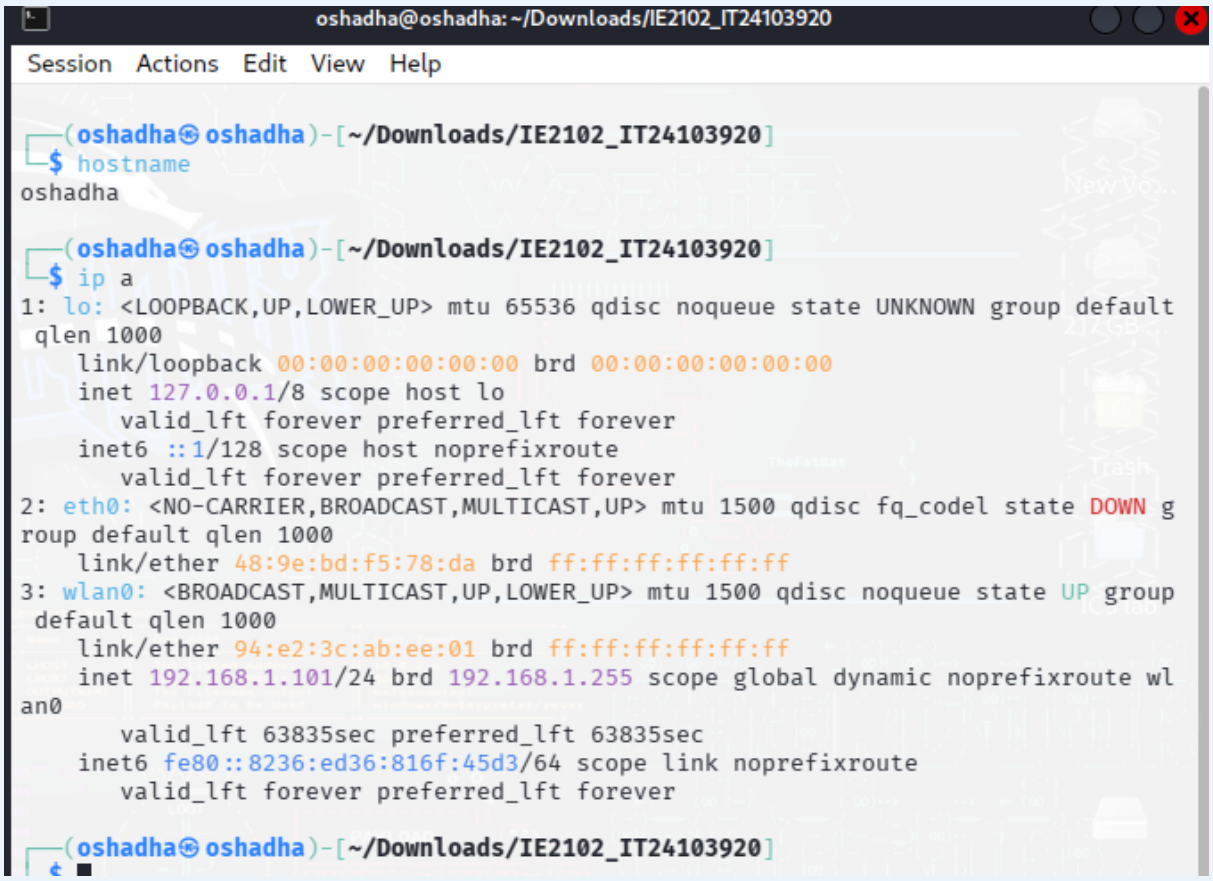
### 3b. Network Configuration

Command used: ip a

*This screenshot shows all network interfaces and their assigned IP addresses, confirming the machine's network setup for the TCP server to bind on.*

### Screenshot 3b – Network Configuration

*Run: ip a | Screenshot must show IP address of loopback and active interfaces*



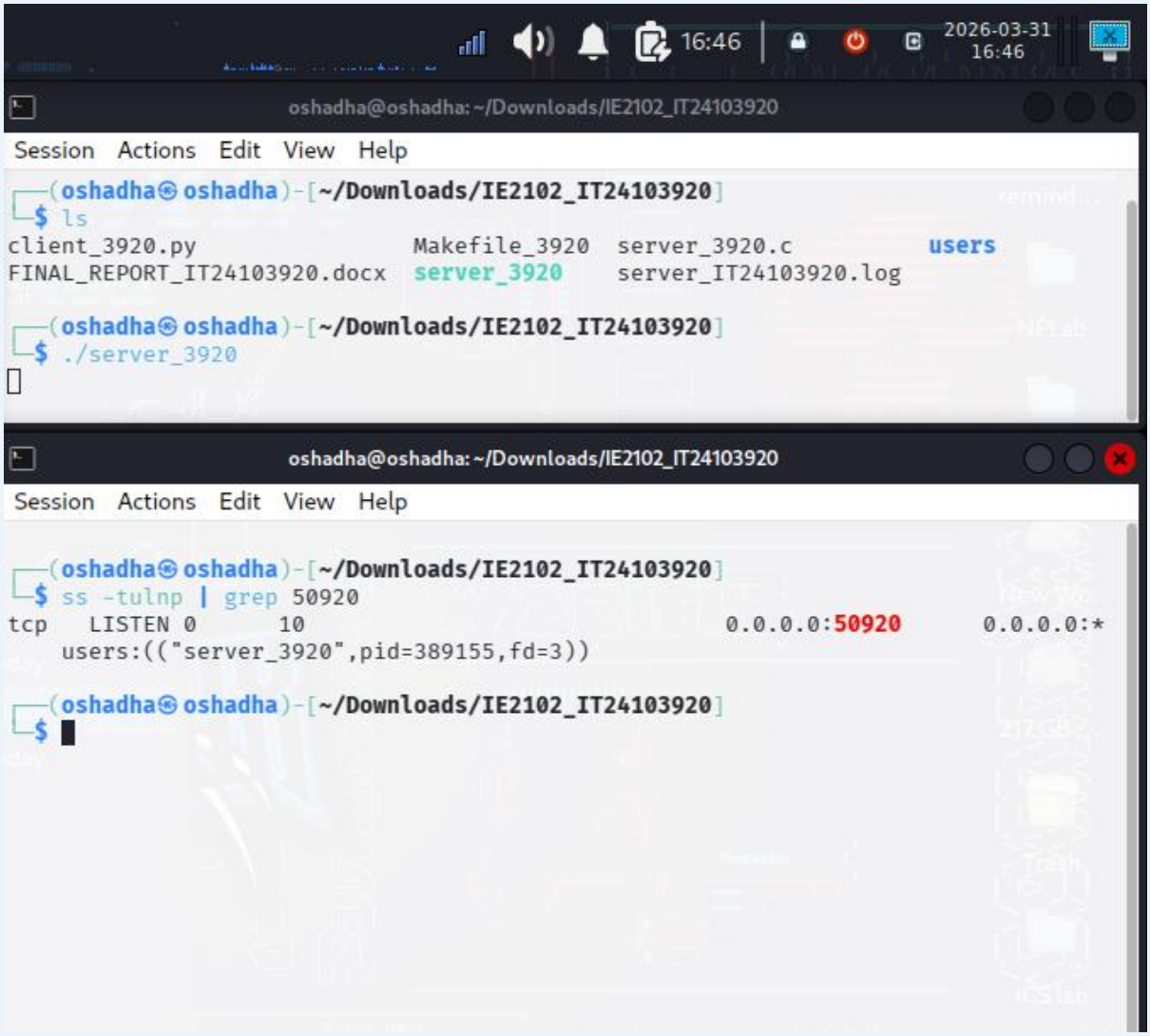
### 3c. Server Running – Port Proof

Commands used: ./server_3920 and ss -tlnp | grep 50920

*This screenshot shows the server process running and port 50920 in LISTEN state. Both the server terminal and the ss/netstat output are captured together.*

### Screenshot 3c – Port 50920 in LISTEN state

*Terminal 1: ./server_3920 | Terminal 2: ss -tlnp | grep 50920*



### 3d. Successful REGISTER and LOGIN

Commands used: REGISTER alice pass123 and LOGIN alice pass123

*This screenshot shows a successful user registration followed by a successful login. The server response includes OK 200 SID:1039 and a session token on login.*

### Screenshot 3d – Successful REGISTER and LOGIN

*Client: REGISTER oshadha otk123 → OK 200 | LOGIN oshadha otk123 → OK 200 TOKEN:...*

```
oshadha@oshadha: ~/Downloads/IE2102_IT24103920
Session  Actions  Edit  View  Help
┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ls
client_3920.py                Makefile_3920  server_3920.c          users
FINAL_REPORT_IT24103920.docx  server_3920    server_IT24103920.log

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ./server_3920
```

```
oshadha@oshadha: ~/Downloads/IE2102_IT24103920
Session  Actions  Edit  View  Help

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ss -tulnp | grep 50920
tcp   LISTEN 0      10                          0.0.0.0:50920      0.0.0.0:*
    users:(("server_3920",pid=389155,fd=3))

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ python3 client_3920.py
>> REGISTER oshadha otk123
OK 200 SID:1039 Registered

>> LOGIN oshadha otk123
OK 200 SID:1039 TOKEN:17749559061804289383

>>
```

```
oshadha@oshadha: ~/Downloads/IE2102_IT24103920
Session  Actions  Edit  View  Help
┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ls
client_3920.py                Makefile_3920  server_3920.c          users
FINAL_REPORT_IT24103920.docx  server_3920    server_IT24103920.log

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ./server_3920
```

```
oshadha@oshadha: ~/Downloads/IE2102_IT24103920
Session  Actions  Edit  View  Help

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ss -tulnp | grep 50920
tcp   LISTEN 0      10                          0.0.0.0:50920      0.0.0.0:*
    users:(("server_3920",pid=389155,fd=3))

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ python3 client_3920.py
>> REGISTER oshadha otk123
OK 200 SID:1039 Registered

>>
```

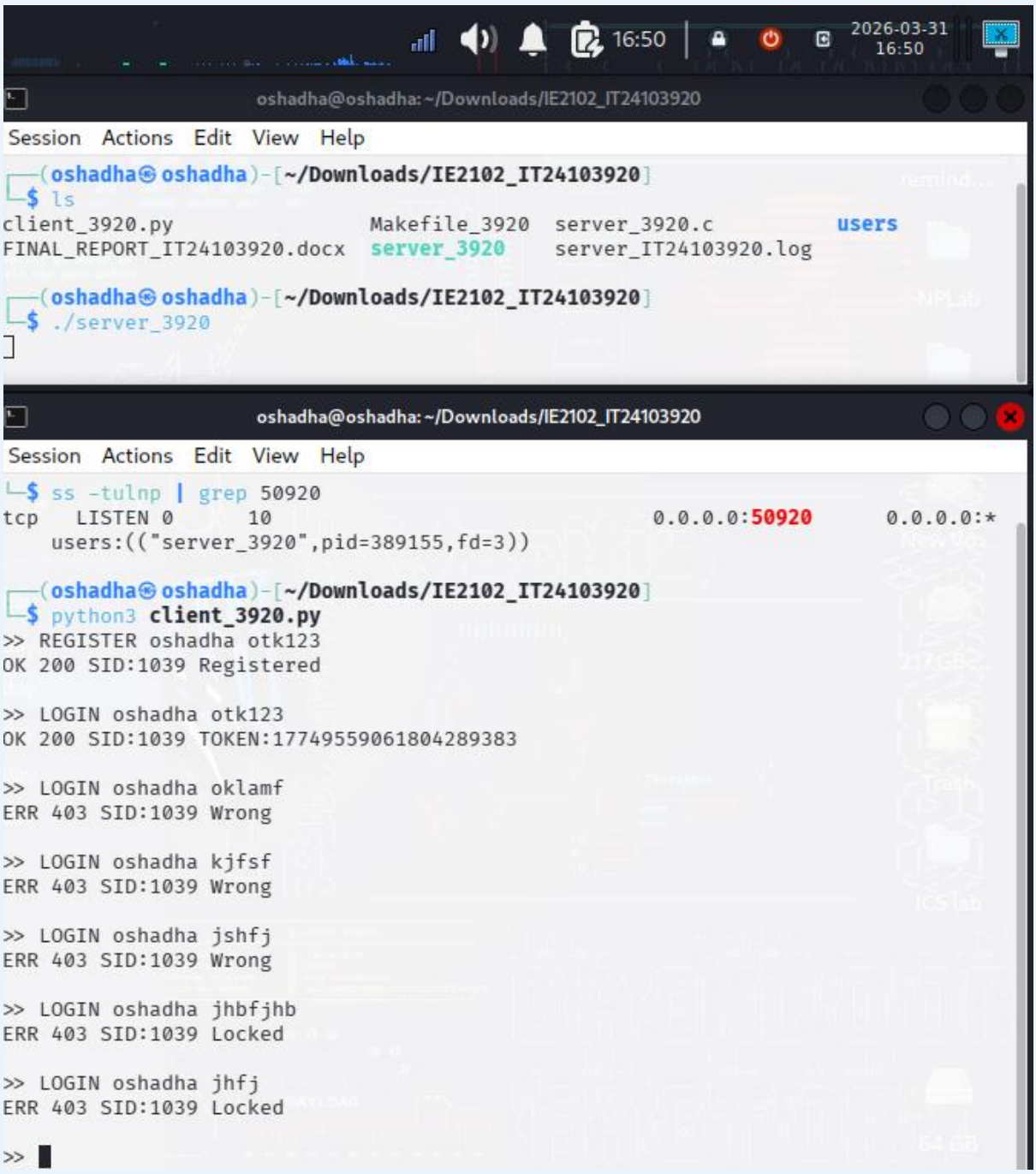
### 3e. Failed Login and Lockout Proof

Commands used: LOGIN alice wrongpass (× 3 attempts)

*This screenshot demonstrates the brute-force lockout protection. After 3 consecutive failed login attempts with an incorrect password, the server returns ERR 403 Locked.*

### Screenshot 3e – Failed Login × 3 and ERR 403 Locked

*ERR 403 | × 3 times → ERR 403 SID:1039 Locked*



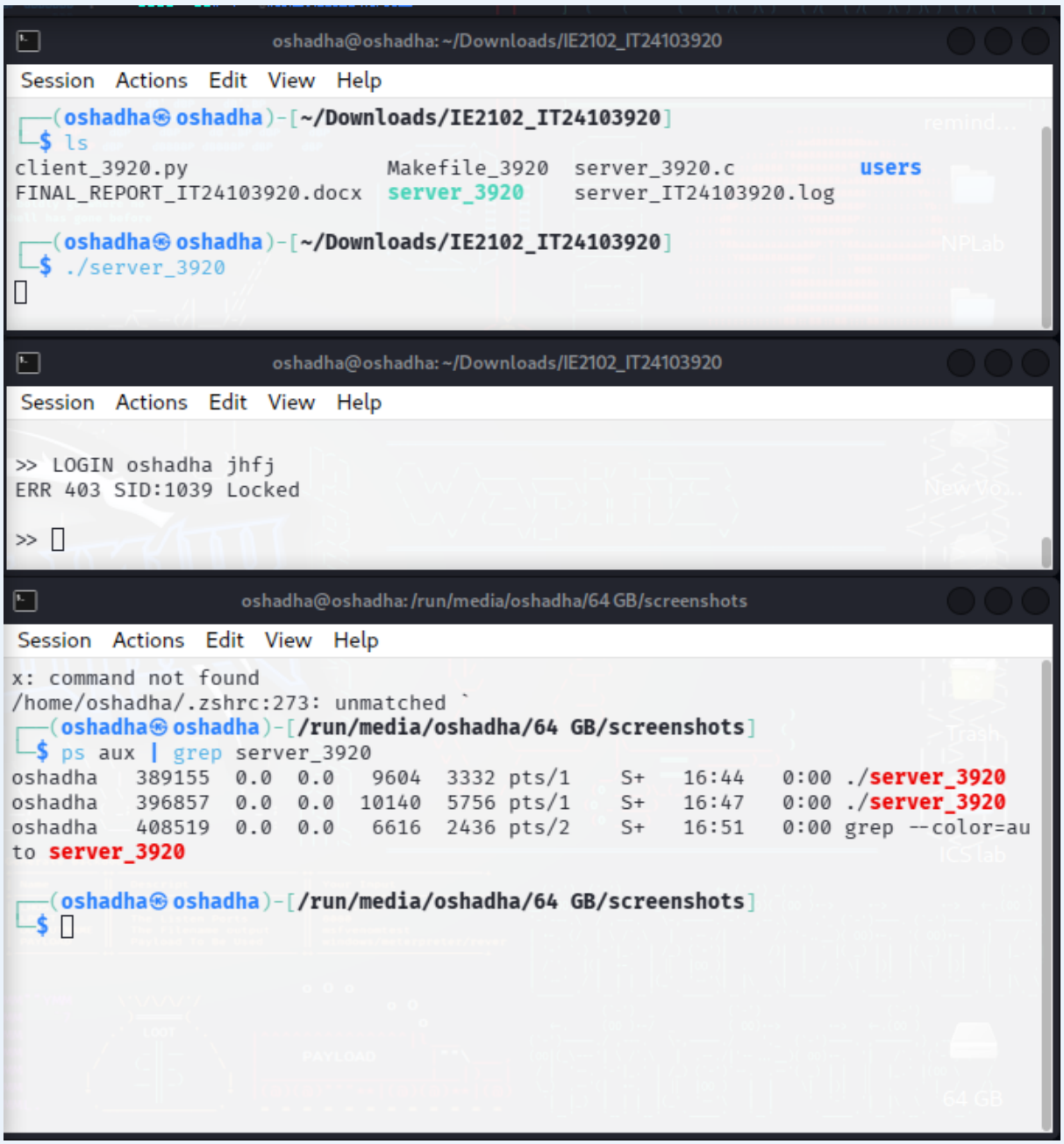
### 3f. Multiple Forked Processes (ps output)

Commands used: Open 3+ client terminals simultaneously, then: ps aux | grep server_3920

*This screenshot shows multiple child processes spawned by the server using fork(). Each connected client results in a separate child process visible in the ps output.*

### Screenshot 3f – Multiple Forked Child Processes

*3+ clients connected simultaneously | ps aux | grep server_3920 → shows multiple PIDs*



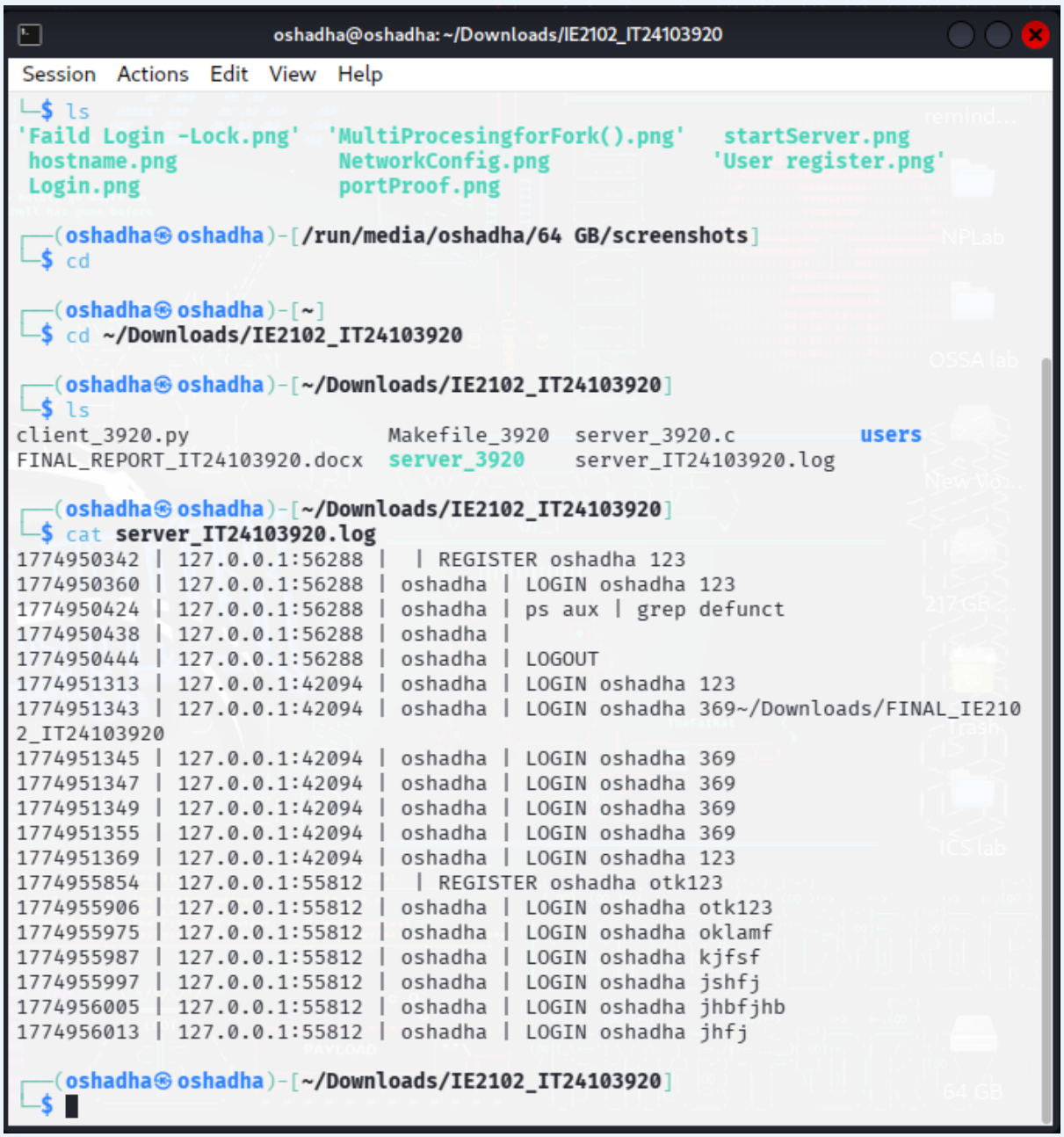
### 3g. Log File Entries

Command used: cat server_IT24103920.log

*This screenshot shows the contents of the audit log file. Each entry contains a Unix timestamp, client IP:port, username (if logged in), and the command that was executed.*

### Screenshot 3g – Audit Log File Contents

*cat server_IT24103920.log | Must show: timestamp | IP:port | username | command*



### 3h. Proper Child Cleanup – No Zombie Processes

Command used: ps aux | grep -E 'defunct|server_3920'

*This screenshot confirms that after clients disconnect, child processes are properly reaped by the SIGCHLD handler using waitpid(). No defunct or zombie processes appear in the ps output.*

### Screenshot 3h – No Zombie Processes

*Disconnect all clients | ps aux | grep -E 'defunct|server_3920' → No [defunct] entries*

oshadha@oshadha: ~/Downloads/IE2102_IT24103920

Session Actions Edit View Help

```
1774950438 | 127.0.0.1:56288 | oshadha |
1774950444 | 127.0.0.1:56288 | oshadha | LOGOUT
1774951313 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 123
1774951343 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369~/Downloads/FINAL_IE210
2_IT24103920
1774951345 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951347 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951349 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951355 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951369 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 123
1774955854 | 127.0.0.1:55812 |  | REGISTER oshadha otk123
1774955906 | 127.0.0.1:55812 | oshadha | LOGIN oshadha otk123
1774955975 | 127.0.0.1:55812 | oshadha | LOGIN oshadha oklamf
1774955987 | 127.0.0.1:55812 | oshadha | LOGIN oshadha kjfsf
1774955997 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jshfj
1774956005 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jhbfjhb
1774956013 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jhfj

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ps aux | grep defunct
oshadha    80297  0.0  0.0      0     0 ?        Z    11:47   0:00 [sd_espeak-ng-
mb] <defunct>
oshadha   415159  0.0  0.0   6616  2364 pts/2    S+   16:55   0:00 grep --color=a
uto defunct

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ps aux | grep defunct
oshadha    80297  0.0  0.0      0     0 ?        Z    11:47   0:00 [sd_espeak-ng-
mb] <defunct>
oshadha   415495  0.0  0.0   6616  2408 pts/2    S+   16:56   0:00 grep --color=a
uto defunct

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$
```

oshadha@oshadha: ~/Downloads/IE2102_IT24103920

Session Actions Edit View Help

```
1774950438 | 127.0.0.1:56288 | oshadha |
1774950444 | 127.0.0.1:56288 | oshadha | LOGOUT
1774951313 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 123
1774951343 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369~/Downloads/FINAL_IE210
2_IT24103920
1774951345 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951347 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951349 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951355 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 369
1774951369 | 127.0.0.1:42094 | oshadha | LOGIN oshadha 123
1774955854 | 127.0.0.1:55812 |  | REGISTER oshadha otk123
1774955906 | 127.0.0.1:55812 | oshadha | LOGIN oshadha otk123
1774955975 | 127.0.0.1:55812 | oshadha | LOGIN oshadha oklamf
1774955987 | 127.0.0.1:55812 | oshadha | LOGIN oshadha kjfsf
1774955997 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jshfj
1774956005 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jhbfjhb
1774956013 | 127.0.0.1:55812 | oshadha | LOGIN oshadha jhfj

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ps aux | grep defunct
oshadha    80297  0.0  0.0      0     0 ?        Z    11:47   0:00 [sd_espeak-ng-
mb] <defunct>
oshadha   415159  0.0  0.0   6616  2364 pts/2    S+   16:55   0:00 grep --color=a
uto defunct

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ps aux | grep defunct
oshadha    80297  0.0  0.0      0     0 ?        Z    11:47   0:00 [sd_espeak-ng-
mb] <defunct>
oshadha   415495  0.0  0.0   6616  2408 pts/2    S+   16:56   0:00 grep --color=a
uto defunct

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$ ps -el | grep Z
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 Z  1000   80297   80281  0  80   0 -     0 -      ?        00:00:00 sd_espeak-n
g-mb

┌──(oshadha㉿oshadha)-[~/Downloads/IE2102_IT24103920]
└─$
```